# Smartphone Specifications and Price

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# Goals & Research Questions

Main goals of the project:

- Compare smartphone performance across brands
- Analyze the effect of additional features on price
- Build a regression model for price explanation

Research questions:

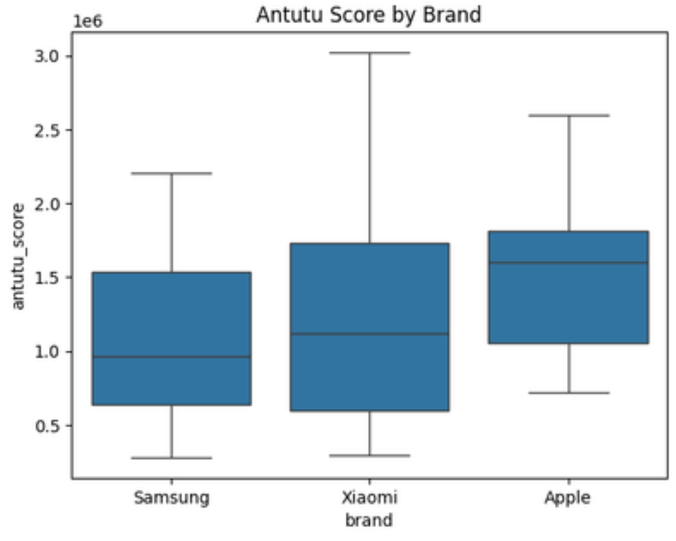
- Do brands differ in performance at the same price level?
- Does wireless charging increase price in budget phones?
- Which specs influence price the most?

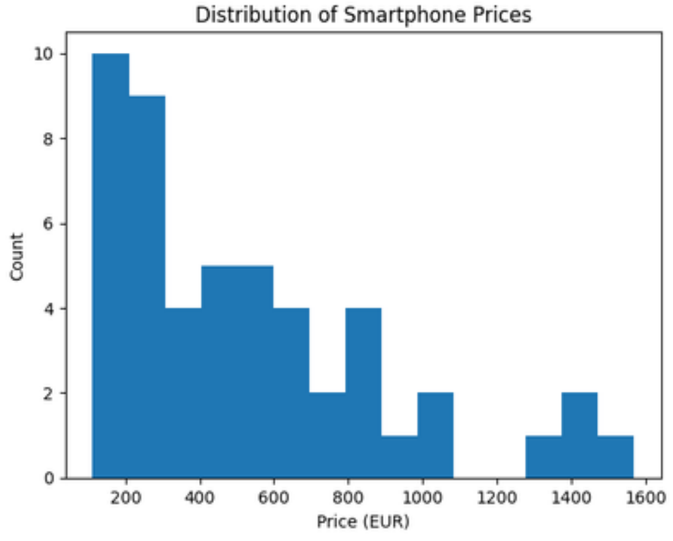
# Data Description

The dataset was collected manually from various sources. It contains 50 models of the most popular phones of 3 brands from 2021 to 2025. It includes information about the characteristics and price of smartphones.

| | model | brand | price_eur | price_segment | release_year | ram | storage | battery | charging_speed | wireless_charging | main_camera | weight | ppi_density | antutu_score |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | Galaxy S21 Ultra | Samsung | 232 | low | 2021 | 12 | 128 | 5000 | 25 | yes | 108 | 227 | 515 | 657150 |
| 1 | Redmi Note 10 Pro | Xiaomi | 187 | low | 2021 | 6 | 128 | 5020 | 33 | no | 108 | 193 | 395 | 295442 |
| 2 | Poco F3 | Xiaomi | 227 | low | 2021 | 8 | 256 | 4520 | 33 | no | 48 | 196 | 395 | 631850 |
| 3 | iPhone 13 Pro Max | Apple | 436 | mid | 2021 | 6 | 128 | 4352 | 23 | yes | 12 | 240 | 458 | 801691 |
| 4 | Xiaomi 12 Pro | Xiaomi | 331 | mid | 2021 | 12 | 256 | 4600 | 120 | yes | 50 | 204 | 521 | 985226 |

# Exploratory Data Analysis



Boxplot
Apple shows higher median performance
Xiaomi has larger variability
Samsung performance is more concentrated

Histogram
Prices from 112 to 1569 EUR
Most phones are cheap or mid-range
Few very expensive models

Scatter Plot
Positive relationship between price and performance
Higher Antutu → higher price
Relationship is not perfectly linear

# One-way ANOVA - Normality Check

```
Normality test for each brand group:

Group: brand = Apple
  W = 0.9748, p = 0.9322
  Fail to reject H0: approximately normal

Group: brand = Samsung
  W = 0.9532, p = 0.6853
  Fail to reject H0: approximately normal

Group: brand = Xiaomi
  W = 0.9748, p = 0.9306
  Fail to reject H0: approximately normal
```

Before applying One-way ANOVA, the normality of the Antutu scores was checked for each brand group using the Shapiro–Wilk test.
The results showed that the performance scores for Apple, Samsung, and Xiaomi were approximately normally distributed.

# Why I chose this test

In the premium segment, smartphones have similar high prices, but brands use different strategies.
This analysis was chosen to check whether brand choice affects performance when price is already high.

# One-way ANOVA - Hypotheses and Results

Null Hypothesis: Mean Antutu scores are equal across brands in the upper price segment.
Alternative Hypothesis: At least one brand has a different mean performance.

```
One-way ANOVA results:
  F-statistic = 0.7141
  p-value = 0.4983
  No significant difference between brands (fail to reject H0)
```

## Conclusion

There is no statistically significant difference in average performance between brands in the premium segment.
This suggests that high price guarantees similar performance regardless of brand, and brand choice is driven more by ecosystem and preferences.

# Welch t-test - Normality Check

Before performing the t-test, the normality of smartphone prices was checked separately for devices with and without wireless charging using the Shapiro–Wilk test.
Both groups were approximately normally distributed.

```
Group: Without wireless charging
  W = 0.9620, p = 0.7563
  Fail to reject H0: approximately normal

Group: With wireless charging
  W = 0.9761, p = 0.9130
  Fail to reject H0: approximately normal
```

# Why I chose this test

I used this test to see if wirless charging is still seen as a premium feature and increases price in the low-price segment or it has become a standard feature that no longer affects pricing of smartphones.

# Welch t-test - Hypotheses and Results

Null hypothesis: Low-price smartphones with and without wireless charging have the same average price.
Alternative hypothesis: There is a difference in average price between these two groups.

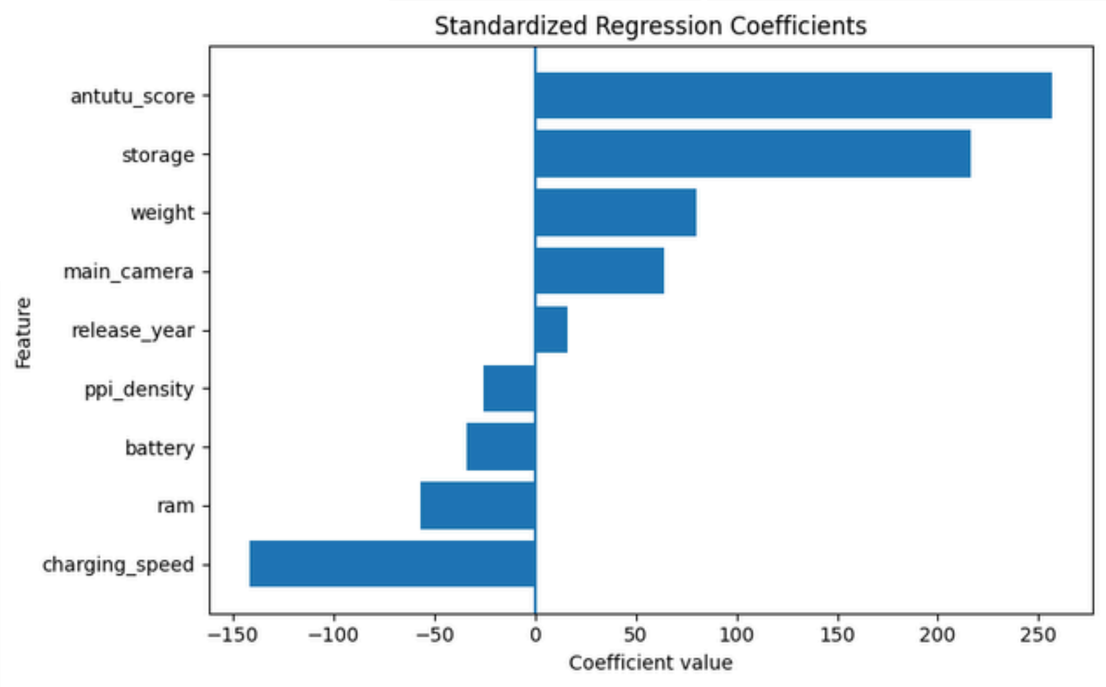
```
Welch two-sample t-test results:
  t-statistic = 1.0315
  p-value = 0.3404
  No significant difference in mean price
```

## Conclusion

Wireless charging does not significantly affect prices in the budget segment.
This indicates that the feature has lost its premium status in low-cost smartphones.

# Multiple Linear Regression

Smartphone prices are often different even when phones have similar technical specifications.
I used Multiple Linear Regression to find which smartphone features really affect the price.




$R^2 = 0.118$
MAE ≈ 219 EUR

Performance (Antutu score) and storage have the strongest positive impact on price.
Charging speed has little or no influence.

# Conclusion

Smartphones with similar high prices have almost the same performance, even if they are from different brands.

Wireless charging does not make low-price smartphones more expensive.

Performance affects the price, but brand image and how people see the brand are also very important.

Overall, we have seen that smartphone prices depend on technical features, but brand image and consumer perception are also very important.